

Implied-Volatility Dynamics at Intraday Reversals

An event atlas of QQQ zero-day-to-expiry options, 2022–2024

Working paper · 16 September 2026 · Descriptive study, not a strategy test

ABSTRACT. At intraday reversals in QQQ, the direction of zero-day implied volatility into the level separates strikes that held from strikes that were run through; the size and composition of dealer positioning at the strike does not. Using 665 trading days of one-minute zero-day-to-expiry (oDTE) QQQ option chains (January 2022 to December 2024) and one-minute Nasdaq-100 futures bars, we identified 1,681 price turns that were followed by a swing of at least 1.97 index points and 371 cases in which price crossed one of the day's three heaviest gamma strikes and continued. For each event we reconstructed the option ladder as it stood one minute earlier: open interest by side, dealer gamma, vega, charm and vanna exposure, per-strike implied volatility with its 30-minute change, and quotes. Hypotheses were formed by reading a stratified sample of 102 events and then checked against the full population. Open-interest walls sat within one point of the event strike at 34 to 38 percent of turns and 38 to 39 percent of run-throughs. At-the-money implied volatility had risen by more than one volatility point over the prior 30 minutes into 53 percent of bottoms but only 17 percent of downside run-throughs, a contrast that held in each of the three years. Implied volatility had fallen by more than one point into 67 percent of upside run-throughs. At 45 percent of bottoms, implied volatility had already retreated more than one point from its 60-minute peak before the price low printed. The study is descriptive and event-conditioned; it characterizes what reversals looked like and does not constitute a forecast test.

1. Introduction

The question is narrow: at the moments when intraday price reversed, what did the zero-day option market look like, and how did that differ from the moments when price reached an apparently similar strike and kept going?

Most work on option positioning and intraday price asks a forecasting question. A signal is defined in advance, a rule fires, and the rule's trades are scored. That design answers whether a particular rule pays. It does not answer what the market actually looked like at the turns, and a rule built on a wrong description of the turns will test as noise regardless of how carefully it is scored. This study inverts the order. Events are selected on the realized price path, the option data at each event is described, and only then are the descriptions compared across event classes.

The comparison class matters. Falling price and rising implied volatility are correlated in equities generally, so “implied volatility rose into the low” is uninformative on its own. The informative comparison is with the run-through: an occasion on which price was also falling into a heavily positioned strike, but did not turn. Any feature that separates the two classes is a candidate mechanism; any feature that does not, however plausible, is not.

Three features of the design follow from this. First, no parameter was chosen by reference to outcomes; the windows and thresholds used in the descriptive statistics were fixed before the population checks and are stated in Section 3. Second, the study is confined to zero-day contracts,

which are the contracts whose hedging is most concentrated within the session. Third, the study produces descriptions and their frequencies, not a trading rule; Section 7 states what a forward test of the description would look like.

2. Data

Two sources were joined by minute over 665 trading days between 3 January 2022 and 31 December 2024. Seven days were excluded because the futures bars for the regular session were missing.

Zero-day option chain. For each trading day, one record per minute, strike and side (call or put) for every QQQ option expiring that day, from 09:31 to 15:59 ET. Each record carries bid, ask and mid, the open interest recorded before the open, a per-strike implied volatility, the forward price implied by put-call parity, and time to expiry measured as minutes to the 16:00 ET close over a 365-day year. The implied volatility is a single value per strike shared by the call and the put, so no put-versus-call skew within a strike is observable. Delta, gamma and the per-strike gamma exposure in the file were verified to reproduce the Black (1976) forward model with a zero rate to displayed precision; charm and vanna were computed from the same inputs with that model. Coverage grew with the product: roughly 130 strikes per minute in 2022, 165 in mid-2024, and 326 on the December 2024 expiry.

Futures bars. One-minute open, high, low and close of the front-month Nasdaq-100 futures contract, timestamped in Eastern Time. These provide the intraday extremes; the option-implied forward is a mid-based series and has no wicks.

Conversion. Futures bars were expressed in QQQ terms by dividing by a rolling 30-minute median of the ratio of futures close to option-implied forward, lagged one minute so that no bar is converted with a ratio that uses its own close. Residual basis noise is on the order of 0.1 to 0.2 index points.

Known defects. On some days every strike in the chain carries a constant fractional offset of about 0.22 points (for example 395.78 for 396), consistent with a distribution adjustment applied at the source; these were restored to whole strikes. Open interest is a single pre-open snapshot, so intraday changes in positioning are not observable; intraday changes in the exposure measures arise only from spot and implied volatility moving over a fixed book.

3. Method

3.1 Event definitions

Turns. A zigzag was run over the converted one-minute highs and lows with a reversal threshold of 1.966 index points. A local extreme qualifies as a turn when the move into it from the previous extreme and the move out of it to the next extreme both reach that threshold. The threshold was chosen as a fixed round figure in futures points (80 points of the Nasdaq-100 micro contract at the prevailing ratio) and was not varied. This produced 1,681 turns: 862 bottoms and 819 tops.

Run-throughs. At 10:00 ET each day the three strikes within one percent of spot carrying the largest absolute gamma exposure were recorded. A run-through is the first minute after 10:00 at which price crossed one of those strikes, provided that price subsequently travelled at least 1.966 points beyond it and no turn printed within 0.5 points of the strike for the rest of the session. This

produced 371 run-throughs: 179 downward through put-heavy strikes and 192 upward through call-heavy strikes.

The two classes are therefore selected by price outcome. This is deliberate: the aim is to describe what the option market looked like at each kind of outcome, not to forecast the outcome.

3.2 Features

For each event the ladder was reconstructed at the minute before the extreme (or the crossing), over the eleven strikes centred on the whole strike nearest the event price:

- open interest, calls and puts;
- dealer gamma exposure and dealer vega exposure per strike, calls positive and puts negative under the standard dealer-long-calls, short-puts convention;
- dealer charm flow, signed so that a positive value means dealers must buy as time passes;
- dealer vanna flow, signed so that a positive value means dealers must buy when implied volatility falls;
- per-strike implied volatility and its change over the prior 30 minutes, both absolute and relative to the change in at-the-money implied volatility;
- option mids and their 30-minute change.

A tape of the event strike's own implied volatility, mids, gamma exposure and vanna flow was recorded at 60, 45, 30, 15, 10, 5 and 1 minutes before the event, alongside spot and at-the-money implied volatility. Session context recorded the time to close, the expected move to the close (spot times at-the-money volatility times the square root of remaining time), and the change in at-the-money volatility over the prior 30 minutes.

3.3 Reading protocol

The protocol had two stages. In the first, a stratified random sample of 102 events (20 turns and up to 14 run-throughs per year, seeded) was printed in full and read event by event, ladder and tape together, to form impressions of what distinguished the classes. In the second, each impression was expressed as a single descriptive statistic and computed over all 2,052 events, split by class and by year. Impressions that did not survive the population check are reported as such. The 30-minute window, the one-volatility-point threshold and the two-times-median definition of a local wall were fixed before the second stage; none was scanned.

4. Results

4.1 Positioning at the event strike does not separate the classes

Turns did not print at strikes. The event price fell within 0.10 points of a whole strike at 22 percent of turns, against 20 percent expected if the location were uniform, and the median distance to the nearest strike was 0.26 points.

Proximity to a local open-interest wall, defined as a strike whose open interest on the defending side is at least twice the ladder median, was the same in both classes.

Class	n	Local wall within 1 pt	Event strike is the ± 3 maximum of gamma exposure	Median signed distance to nearest wall
Turn, bottom	862	38%	21%	1 strike below
Turn, top	819	34%	19%	1 strike above
Run-through, down	179	39%	44% (selected on it)	1 strike below
Run-through, up	192	38%	44% (selected on it)	1 strike above

Table 1. Positioning at the event strike. Chance for the gamma-maximum column is about 14 percent.

Both turns and run-throughs sat about one strike in front of the nearest wall on the defending side. The wall marks where price interacts with positioning; it does not mark which way the interaction resolves.

Two further exposure measures added nothing on zero-day contracts. Vega exposure produced the same map as gamma exposure, because both are proportional to the same density term when time to expiry is hours. Charm flow was positive at the event strike at 63 to 69 percent of bottoms and at 65 to 80 percent of downside run-throughs; its sign is set by the strike's position relative to spot and does not vary with outcome. Vanna flow at the event strike was close to zero in every class, as expected at the money.

4.2 Implied-volatility direction into the event

The direction of at-the-money implied volatility over the 30 minutes before the event separated the classes strongly.

Class	n	IV rose by more than 1 vol pt	IV fell by more than 1 vol pt	Median change (vol pts)
Turn, bottom	862	53%	23%	+1.16
Run-through, down	179	17%	53%	-1.18
Turn, top	819	27%	51%	-1.07
Run-through, up	192	7%	67%	-2.05

Table 2. Change in at-the-money oDTE implied volatility over the 30 minutes before the event.

The bottom versus downside run-through contrast held in each year: rising into 54 versus 17 percent in 2022, 41 versus 15 percent in 2023, and 57 versus 19 percent in 2024. The upside contrast also held each year (falling into 74, 64 and 66 percent of run-throughs against 53, 51 and 48 percent of tops).

The same asymmetry appeared within tops. Tops reached on rising implied volatility were the sharp squeeze-type extremes in the sample (rises of 15 to 40 volatility points in 30 minutes on several late-session examples). The size of the subsequent move did not depend on the direction of implied volatility into the extreme: the median post-turn leg was 0.83 to 1.03 expected moves in every subclass.

4.3 Implied volatility retreats before price does

At 45 percent of bottoms, at-the-money implied volatility had already fallen more than one volatility point from its 60-minute peak by the minute before the price low; the median retreat was 0.8 points and the upper quartile 3.1 points. The mirror figure at tops was 43 percent. In the read sample this appeared as the final push into the extreme printing on implied volatility that was already rolling over.

4.4 Skew shape into the event

Over the same 30-minute window, the change in implied volatility at strikes two to five points below the event was compared with the change at the money.

Class	Median wing-minus-ATM change (vol pts)	Share with wing decompressing by more than 1 pt
Turn, bottom	-1.45	57%
Run- through, down	-0.48	40%
Turn, top	+1.82	8%
Run- through, up	+1.26	8%

Table 3. Downside-wing implied volatility change relative to the at-the-money change, prior 30 minutes.

Into a bottom that held, the downside wing rose less than the at-the-money strike: the repricing was concentrated in the front strikes and the tail was not being re-marked. Into a downside run-through, the wing stayed relatively bid. The upside separation was weaker.

4.5 Large late-session pins attract price; the extreme prints in front of them

In the read sample, the largest late-session concentrations of call open interest and positive charm flow above spot drew price toward them, and the top printed one to four points short of the strike. Two examples: a top at 361.33 with 47 minutes remaining under a 365 strike carrying 36,039 calls, and a top at 472.45 with 23 minutes remaining under a 473 strike carrying 31,986 calls. The strike

itself was not reached in either case. This is a sample observation and was not reduced to a population statistic.

4.6 Time of day

Turns were concentrated in the 10:00 hour (433 of 1,681, or 26 percent). Run-throughs were earlier events (median 345 minutes to the close, against 235 for turns), consistent with their definition from the 10:00 ladder.

5. Interpretation

The results are consistent with a simple mechanical reading. Zero-day positioning determines where price will interact with dealer hedging, and the ladder locates those strikes well. Whether the interaction produces a turn depends on whether there is a bid for protection to exhaust.

A bottom that holds is reached with implied volatility rising: participants are paying for downside protection into the low, dealers who sell that protection hedge by buying the underlying as their short puts gain delta, and the low prints when the bid for protection stops, which the data show happening before the price low in almost half the cases. A downside run-through is reached with implied volatility falling: the decline is orderly, there is no rush for protection, no exhaustion point, and the strike offers nothing to lean on regardless of how much open interest sits at it.

The upside is the mirror with one addition. A rally on falling implied volatility carries the vanna tailwind, in which falling volatility shrinks the delta of the puts dealers are short and forces them to buy; such rallies passed through call-heavy strikes two-thirds of the time. A rally on rising implied volatility is a squeeze, in which calls are being bought into strength; such rallies produced the sharp tops in the sample.

The skew result refines this. Into a real low the repricing is in the front strikes, not the tail: the market is paying for immediate protection, not re-marking the probability of a large move. When the wing stays relatively bid into the decline, the market is pricing continuation, and continuation is what follows.

What the results do not show should be stated as plainly. They do not show that rising implied volatility predicts a bottom; they show that bottoms, once identified from the price path, were usually preceded by rising implied volatility, whereas run-throughs were not. The population of all minutes on which implied volatility was rising into a put-heavy strike is larger than either class and was not examined. Nor do the results identify a level at which the turn will print; they indicate that the turn prints in front of the wall rather than at it, on both sides, and that the strike itself is a poor target for a resting order.

6. Limitations and the question of overfitting

Overfitting in the usual sense did not occur. No parameter was optimized against outcomes. The swing threshold, the 30-minute window, the one-volatility-point cutoff, the two-times-median wall definition and the 60-minute tape span were each set once, before the population statistics were computed, and none was varied to improve a result. There is no fitted model, no coefficient, and no rule whose performance was maximized.

The weaker form of the risk is real and should be named. Hypotheses were formed by reading 102 events drawn from the same 2,052 that were then used to check them. That is in-sample description: the sample and the population overlap, and a reader will notice patterns that happen to be present in the sample. Three things limit the damage. The population is 20 times the sample, so a feature present only in the sample would be diluted rather than confirmed. The main contrast was split by year and held in each of three regimes with different volatility levels. And the features that did not survive the check (charm sign, wall proximity, gamma rank) are reported alongside those that did. The correct discipline is nonetheless a forward test on data not yet examined, described in Section 7.

Selection on outcome. Events were defined by the realized price path. This is appropriate for description and inappropriate for inference about prediction: the statistics are conditional on the event having occurred. The base rate of the volatility signature across all minutes is not measured here.

Class asymmetry. Run-throughs were drawn only from the day's three heaviest gamma strikes at 10:00, so that class is by construction a set of heavily positioned strikes reached early in the session. Turns were drawn from the whole session. The time-of-day contrast in Section 4.6 is therefore partly definitional, and the positioning comparison in Section 4.1 is biased toward finding positioning at run-throughs. That bias runs against the paper's conclusion, not for it.

Confounding with the spot-volatility relation. Implied volatility tends to rise when price falls. The rising-into-bottoms statistic would be uninformative on its own; the finding is that downside run-throughs, on which price was also falling, did not show it.

Data. Positioning is zero-day only; the multi-expiry book is not observed. Implied volatility is a single parity-consistent value per strike, so put-call skew within a strike is not observed. Open interest is a pre-open snapshot. The futures-to-QQQ conversion carries basis noise of roughly 0.1 to 0.2 points. Trade volume is absent.

7. Suggested forward test and reproducibility

The description can be turned into a falsifiable statement without a fitted parameter. Fix in advance: a level definition (a strike with at least twice the ladder-median open interest on the defending side, within one expected move of spot), the 30-minute window, the one-point cutoff, and the outcome definition used here. Then, on a year of data not used in this study, record every minute at which price first came within 0.5 points of such a strike from the appropriate side, classify the preceding implied-volatility change, and tabulate outcomes by that classification. The prediction is a difference in hold rate of the order seen here (roughly three to one in the share of rising-volatility approaches between the two outcome classes). One-minute option quote data for the same product from a later year would serve; the signature requires only quotes and an underlying price series, not open interest.

A second test targets the timing result: at each such approach, whether implied volatility has retreated more than one point from its trailing 60-minute peak, and whether that state precedes the outcome.

Reproducibility. The event construction, ladder features and tape are produced by a single script over the two data sources; the descriptive statistics in Section 4 are computed from its output by short

scripts. The stratified sample is seeded. All thresholds appear in Section 3.

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